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University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination — Semester II - UCSC AY20 [held in March/ April 2024]

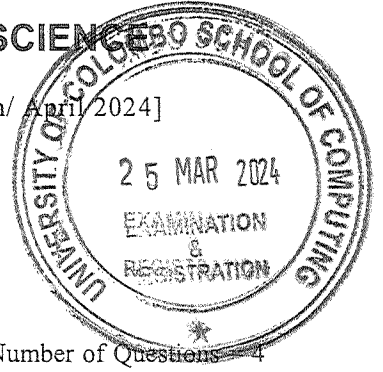
SCS 1211 – Mathematical Methods - I

(Two (2) Hours)

Answer ALL questions

Number of Pages = 4

Number of Questions = 4



Important Instructions to candidates:

- I. Students should answer in the medium of **English language only** using the answer sheet provided.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of the answer sheet.
- IV. This paper consists of 4 questions on 04 pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.
- VII. **Non-Programmable** calculators are **allowed**.

1)

a) Let $A = \begin{bmatrix} 0 & 2 & 3 & 7 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 4 & 8 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

- i. Find the **row reduced echelon form** of the matrix A .

[7 Marks]

- ii. Hence find the general solution to the following system of homogeneous equations:

$$\begin{aligned} 2x_2 + 3x_3 + 7x_4 &= 0 \\ x_1 + x_2 + x_3 + x_4 &= 0 \\ x_1 + 3x_2 + 4x_3 + 8x_4 &= 0 \\ x_4 &= 0. \end{aligned}$$

[5 Marks]

b) Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 2 & -1 & 2 \end{bmatrix}$.

- i. Using the Gauss-Jordan Method, find the inverse of A .

[8 Marks]

- ii. Find the equation of the plane that contains the points $(1,1,1)$, $(1,3,2)$ and $(2,-1,2)$.

[5 Marks]

2)

- a) With the usual notations, define the term **Vector space**.

[4 Marks]

Let V be a vector space over the field F , and let $u, v \in V$, $\alpha \in F$. Prove the following statements:

- i. $u \oplus v = u$ implies $v = \mathbf{0}_v$, where $\mathbf{0}_v$ is the zero vector of V .

[2 Marks]

- ii. $0 \odot u = \mathbf{0}_v$, where 0 is the zero element of the field F .

[2 Marks]

b) Let $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 2 \end{bmatrix}$.

- i. Find the column space of A ($C(A)$), a basis for $C(A)$, and the dimension of $C(A)$.
[6 Marks]
- ii. Find the Null space of A ($N(A)$), a basis for $N(A)$, and the dimension of $N(A)$.
[6 Marks]
- iii. Extend the basis obtained in part (ii) of $N(A)$ to the basis of $\mathbb{R}^4$.
[5 Marks]

3)

- a) Let A be an $m \times n$ real matrix. Show that the vector function $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $Tx = Ax, \forall x \in \mathbb{R}^n$ is a linear transformation.
[3 Marks]
- b) Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation from the vector space $\mathbb{R}^n$ to the vector space $\mathbb{R}^m$. Show that there exists an $m \times n$ real matrix A such that $Tx = Ax, \forall x \in \mathbb{R}^n$.
[6 Marks]
- c) Let T, S be two linear transformations from the vector space $\mathbb{R}^2$ to itself represented by the matrices $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$ respectively.
 - i. Give full geometrical descriptions for each of the transformations represented by A and B .
[4 Marks]
 - ii. Let PQR be the triangle with vertices at the coordinates $P \equiv (1, -1), Q \equiv (3, -1)$ and $R \equiv (3, 3)$. The triangle PQR is first transformed by T and then by S , producing the triangle $P'Q'R'$. Find the single matrix that represents this composite transformation, determine the coordinates of the vertices of the triangle $P'Q'R'$, and the area of the triangle $P'Q'R'$.
[6 Marks]
 - iii. The triangle $P'Q'R'$ is then reflected in the straight line with equation $x + y = 0$ to give the triangle $P''Q''R''$. Find the single matrix that maps $P''Q''R''$ back onto PQR .
[6 Marks]

4)

- a) Using the properties of a determinant, find the determinant of the matrix $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 3 \\ 1 & 1 & 4 \end{bmatrix}$.
[6 Marks]

- b) Let A be a square matrix, and suppose that A has **distinct** eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ with corresponding eigenvectors $x_1, x_2, \dots, x_k$. Prove that $x_1, x_2, \dots, x_k$ are linearly independent.

[4 Marks]

c) Let $A = \begin{bmatrix} 2 & -5 & 0 \\ -5 & -1 & 3 \\ 0 & 3 & -6 \end{bmatrix}$.

- i. Show that 6 is an eigen value of A and find the other two eigenvalues of A .

[5 Marks]

- ii. Find the eigen vectors of unit length corresponding to the eigen values obtained in part (i).

[6 Marks]

- iii. Let P be the square matrix containing the three eigen vectors of unit length obtained in part (ii) as its columns. Show that P is an orthogonal matrix.

[2 Marks]

- iv. Show that $P^{-1}AP$ is a diagonal matrix.

[2 Marks]